



AStar Bootcamp – NTI Global Talent Program

Our Vision

AStar is an NTI Global Talent Program designed to empower the next-generation of young talented engineers and researchers, particularly emerging leaders from the broader Asia-Pacific region and beyond. Through this 12-week intensive program, combining advanced technical training and leadership development, students are expected to master state-of-the-art large language model training techniques as well as work on high-impact research and industry projects. The program culminates in a showcase of top projects at the GenAI Summit in November 2025.

Our Participants

- **Inaugural Cohort Size:** 30-50 students.
- **Profile:** Bachelor's degree or higher in CS or related fields; prior experience in ML/DL.
- **Selection:** Competitive application, technical test, and interview

Our Program

- **Timeline: Aug - mid Nov 2025**
 - Jul 28: public announcement of the program.
 - Aug 9: press conference announcement
 - Aug 11: students take an entrance test followed by individual interviews.
 - Aug 15: announcement of the inaugural cohort
 - Aug 18 - Nov 7: NTI Global Talent program (12 weeks)
 - Nov 15 or 22: GenAI summit (<https://summit.newturing.ai/>).
- **Advanced Technical Foundations**
 - Focus areas: Advanced NLP and Advanced LLM
 - Blended lectures, tutorials, and assignments designed to deepen technical skills in foundation models.
- **Capstone Project**
 - Students work independently for the first 6 weeks, then form teams to collaborate subsequently, guided by technical buddies and advisors.
 - Projects are evaluated on technical novelty and potential impact.

- **Leadership Development:** we would like to nurture the next-generation of leaders. Guest lectures, project reports and presentations together with feedback from advisors will hopefully provide useful coaching to cultivate well-rounded AI leaders.
- **Final deliverables:** code repository, technical presentation, and poster – showcased at the **GenAI Summit 2025 in mid November**.

Week	State-of-the-art LLM Training	Capstone project
1	Review of advanced NLP techniques Goal: recap fundamental knowledge such as attention, tokenization, Transformer, and end-to-end training (e.g., nanoGPT).	Individual Project brainstorming Goal: think critically about interesting and impactful ideas that can be achieved in 12 weeks Deliverables: project proposal (2-4 pages) to be reviewed by technical buddies & program leads.
2	Pre-training Goal: develop in-depth understanding of attention (e.g., Linear Attention, FlashAttention), advanced Transformers (Sparse Transformer - Transformer-XL, MoE) and LLM pretraining. Tutorial: <i>Train a large language model from scratch</i> . Assignment 1 (week 1-2): optimizer state sharding (reference).	
3	Post-training Goal: understand supervised finetuning (SFT) and parameter-efficient methods. Learn how to prepare data, setup training, hyperparameter optimization, perform decoding, and evaluation.	Individual Project prototype Goal: implement the first version of the project proposal utilizing knowledge learned during the first 6 weeks. Deliverables: project presentation presented to technical buddies, program leads, and senior advisors
4	Alignment & Advanced reasoning Goal: understand learning from human feedback (RLHF, DPO) and recent developments for advanced reasoning models (DeepSeek-R1 and GRPO).	
5	Assignment 2 (week 3-5): train math reasoning models with GRPO (reference).	
6	Project prototype presentation (Individual students will merge into teams for collaborative works)	
7	Guest lectures on either or both: • Text diffusion models • Flash attention	Team Project refinement Goal: refine the idea and implementation based on feedback of

8	Advanced Inference <i>Goal: kv-caching, advanced decoding algorithms & serving optimizations.</i>	the prototype. <i>Deliverables:</i> * project mid-term report (2-4 pages) on the latest results * initial code submission <i>To be reviewed by technical buddies & program leads.</i>
9	Agents and Evaluation <i>Goal: understand AI agents and automated benchmarks (BLEURT, COMET, LLM judge)</i>	
10	Guest lectures on either or both: - AI Safety - AI for Drug Discovery	Team Project completion <i>Goal: complete the project with polished results and implementation.</i>
11	Guest lectures on either or both: - Leadership - Research skills	<i>Deliverables:</i> * final project presentation to everyone * project final report (4-8 pages) * final code submission <i>To be graded by technical buddies & program leads.</i>
12	Final project presentation (later followed by blog-post/Camera-ready presentation to post on Medium/HF/NTI website)	

Our Outcomes

- **Capstone Projects:** High-impact research and industry-ready capstone projects. Top teams showcase at GenAI Summit (Nov 2025).
- **Industry & Research Pathways:** Direct talent pipelines to partner companies (e.g., Google DeepMind, NVIDIA, Qualcomm, etc.) & opportunities at leading research labs (e.g., Stanford, NUS etc.).
- **Leadership Growth:** Development of essential leadership skills and access to NTI's global alumni network.